MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **GENERAL ARRANGEMENT**
POWER GENERATION : SOLAR PLANT
MODULE MOUNTING STRUCTURE FOUNDATION LAYOUT

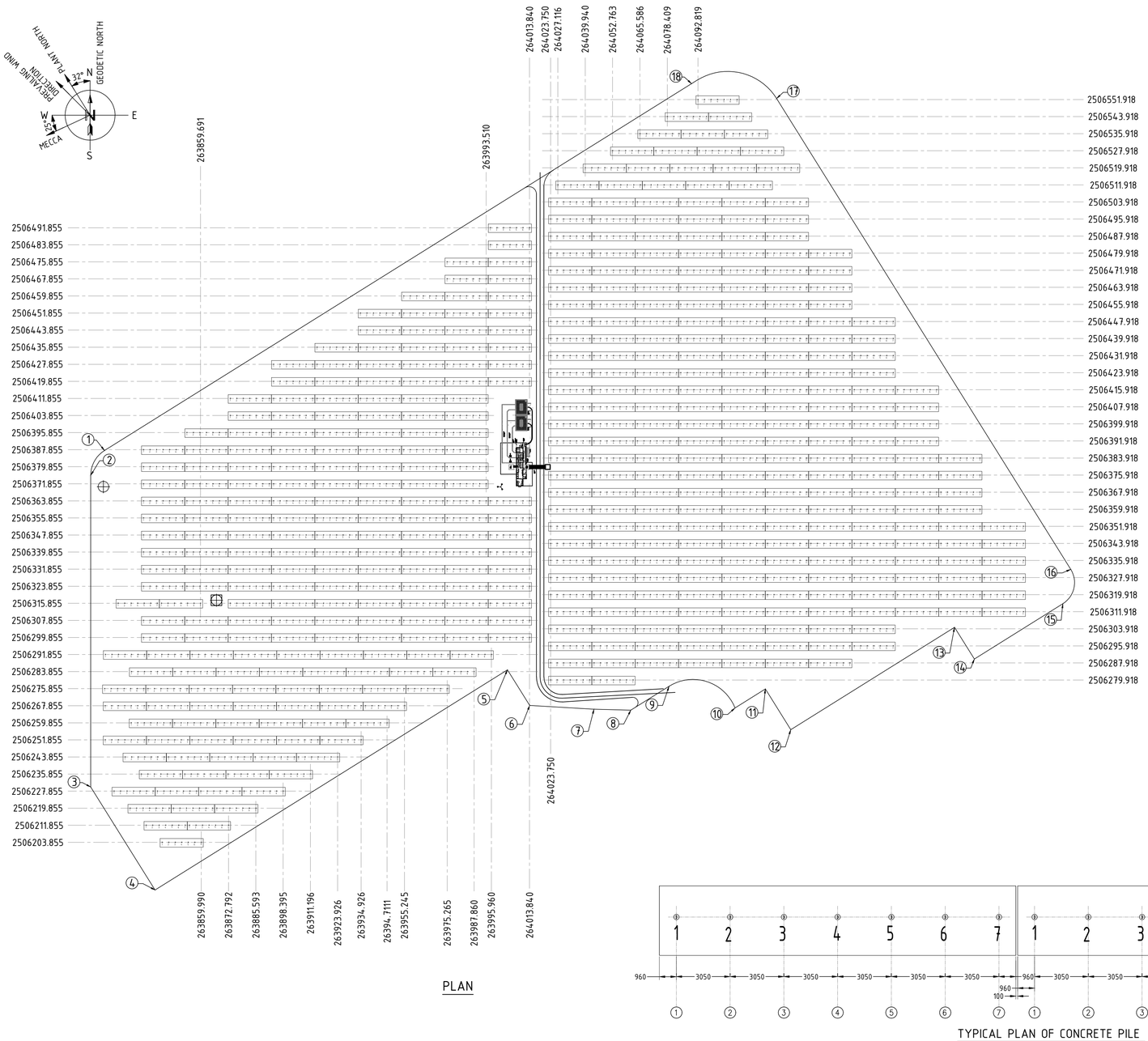
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VISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	27.02.2020	ISSUED FOR REFERENCE	RVJ	AVG	SJM	
C01	03.10.2020	ISSUED FOR CONSTRUCTION	RVJ	AVG	SJM	
C02	17.07.2021	ISSUED FOR CONSTRUCTION	RVJ	AVG	SJM	
D01	09.03.2023	AS BUILT	RVJ	AVG	SJM	

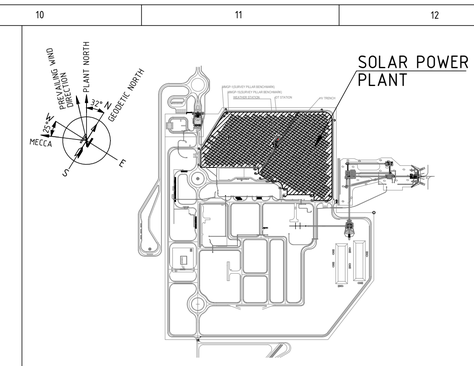
DRAWING NUMBER: MD-216-49PS-EG-CV-GAD-1603 | REV: D01



PLAN

PLOT BOUNDARY CO-ORDINATES

DESCRIPTION	EASTING	NORTHING
1	263814.667	2506387.430
2	263808.227	2506375.812
3	263808.227	2506229.797
4	263838.391	2506181.525
5	264003.506	2506284.700
6	264013.928	2506268.021
7	264044.064	2506266.216
8	264061.313	2506265.792
9	264079.220	2506276.981
10	264110.156	2506266.828
11	264124.377	2506275.714
12	264136.298	2506256.637
13	264213.029	2506304.583
14	264222.302	2506289.742
15	264263.771	2506315.655
16	264267.493	2506331.777
17	264129.630	2506552.403
18	264092.425	2506560.992

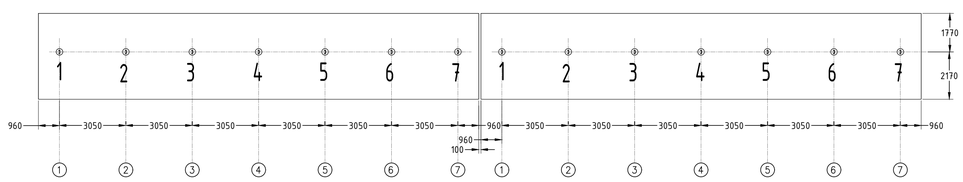


KEY PLAN

LEGEND:

- PLOT BOUNDARY
- IDT STATION
- WEATHER STATION
- MMGP-15 (SURVEY PILLAR BENCHMARK)
- MMGP-1 (SURVEY PILLAR BENCHMARK)
- TABLE

N.G.L. - NATURAL GROUND LEVEL
TYP. - TYPICAL
NTS. - NOT TO SCALE
MMS. - MODULE MOUNTING STRUCTURE
℄ - CENTRE LINE



TYPICAL PLAN OF CONCRETE PILE

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETRES, LEVEL AND CO-ORDINATES ARE IN METRE.
2. IN CASE OF ANY DISCREPANCY IN THE DIMENSION, WRITTEN DIMENSION SHALL BE FOLLOWED
3. ANY DISCREPANCY IN THE DRAWING SHALL BE BROUGHT TO THE NOTICE OF L&T DESIGN TEAM.
4. FOUNDATION SHALL BE BORED CAST IN SITU WITH POST EMBEDDED IN CONCRETE.
5. FINAL PILE LOCATIONS SHALL BE DECIDED IN CONSULTATION WITH ENGINEER-IN-CHARGE AT SITE.
6. SOIL CHARACTERISTICS : SAND WITH SILT AND GRAVEL / GRAVEL WITH SILT AND SAND IS ASSUMED
7. WHILE INSTALLING THE PILE BORE, SOIL PROFILE SHALL BE CHECKED WITH RESPECT TO SOIL PROFILE OBTAINED FROM GEO-TECHNICAL REPORT IN CASE OF ANY MAJOR DIFFERENCE BETWEEN THE TWO,THE SAME SHALL BE REPORTED TO L&T DESIGN OFFICE.
8. COORDINATES SYSTEM SHOWED IS UTM WGS 84 SYSTEM - ZONE 37N

SHEET 1 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	09/03/2023	AS-BUILT	RVJ	AVG	GSK	SJM	
		C02	17/07/2021	ISSUED FOR CONSTRUCTION	RVJ	AVG	GSK	SJM	
MD-216-49PS-EG-CV-GAD-1604	MMS FOUNDATION CROSS SECTION AND DETAILS	C01	03/10/2020	ISSUED FOR CONSTRUCTION	RVJ	AVG	GSK	SJM	
MD-216-49PS-EG-EL-LAD-1002	LAYOUT SOLAR POWER PLANT OVERALL	A01	02/09/2020	ISSUED FOR REFERENCE	RVJ	AVG	GSK	SJM	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__WG02_MTL01__45003 | DOCUMENT ID: OU100688096 | REVISION: 3

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CHECKED	AVG	24/08/2020
ENGINEER	GSK	23/08/2020
APPROVED	SJM	02/09/2020
EMPLOYER		
SCALE	NTS	A1

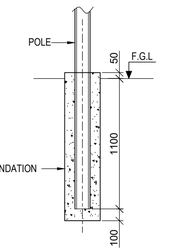
MANSOURAH - MASSARAH GOLD PROJECT
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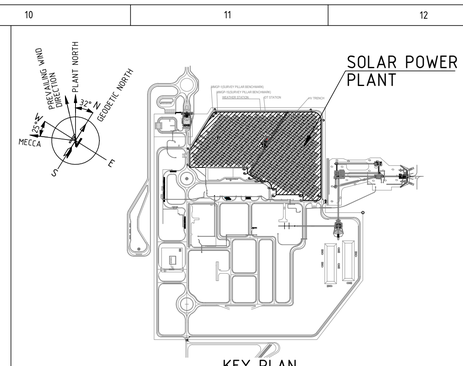
INVERTER FOUNDATION
SCALE 1:25

SECTION X-X
SCALE 1:25

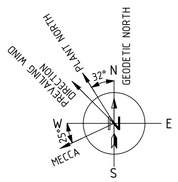
ACB, DBB & ETH FOUNDATION
SCALE 1:25

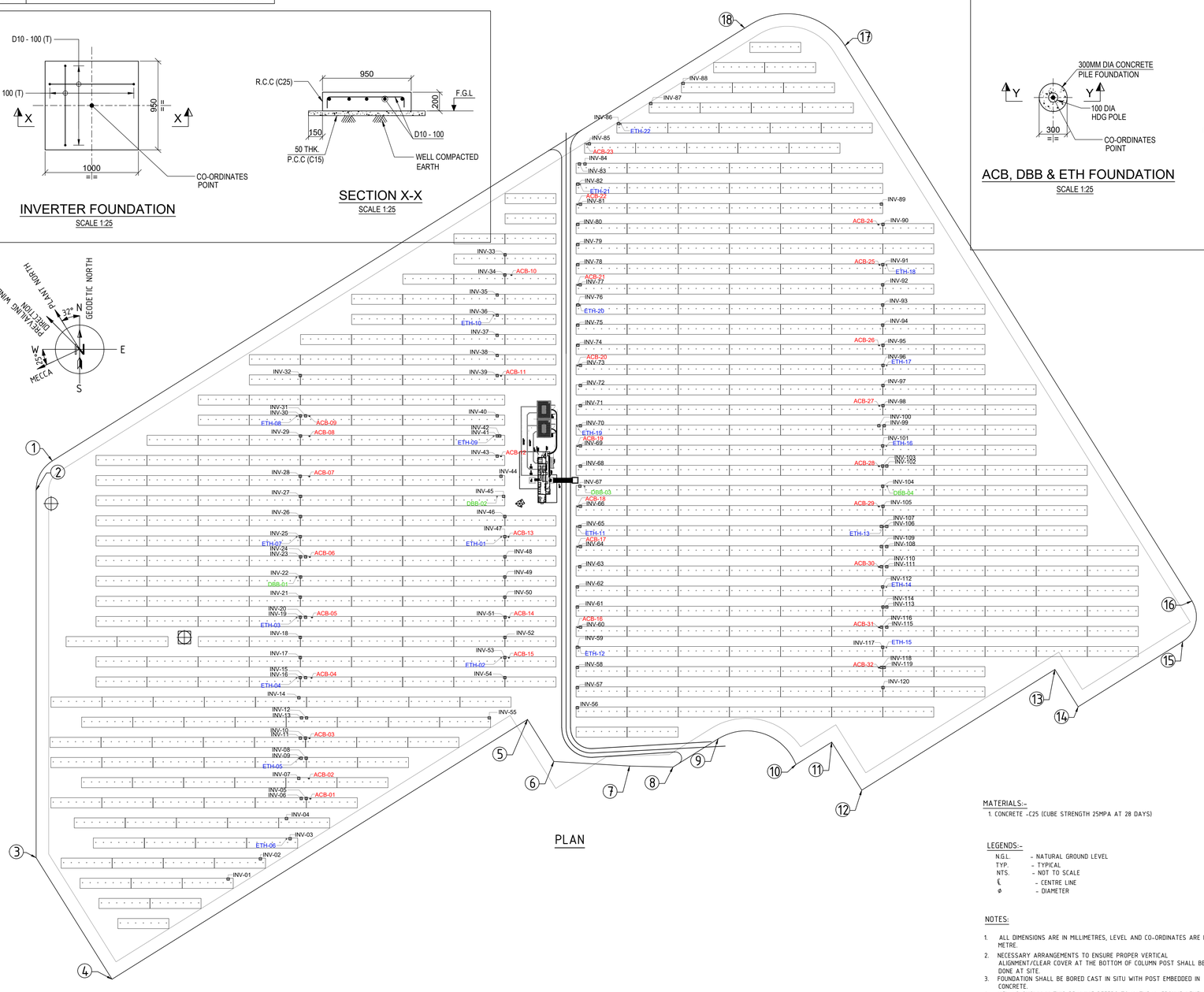


SECTION Y-Y
SCALE 1:25



KEY PLAN





PLAN

MATERIALS:-

1. CONCRETE -C25 (CUBE STRENGTH 25MPA AT 28 DAYS)

LEGENDS:-

N.G.L.	- NATURAL GROUND LEVEL
TYP.	- TYPICAL
NTS.	- NOT TO SCALE
℄	- CENTRE LINE
ø	- DIAMETER

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES, LEVEL AND CO-ORDINATES ARE IN METRE.
2. NECESSARY ARRANGEMENTS TO ENSURE PROPER VERTICAL ALIGNMENT/CLEAR COVER AT THE BOTTOM OF COLUMN POST SHALL BE DONE AT SITE.
3. FOUNDATION SHALL BE BORED CAST IN SITU WITH POST EMBEDDED IN CONCRETE.
4. NGL AS SHOWN IN THIS DRAWING REFERS TO NATURAL GROUND LEVEL AT SITE ANY UNDULATION BETWEEN ADJACENT COLUMN POST TO BE ADJUSTED IN COLUMN HEIGHTS ABOVE NGL BY PV TABLE ERECTION MANUAL.
5. FINAL PILE LOCATIONS SHALL BE DECIDED IN CONSULTATION WITH ENGINEER-IN-CHARGE AT SITE.
6. WHILE INSTALLING THE PILE BORE, SOIL PROFILE SHALL BE CHECKED WITH RESPECT TO SOIL PROFILE OBTAINED FROM GEO-TECHNICAL REPORT. IN CASE OF ANY MAJOR DIFFERENCE BETWEEN THE TWO, THE SAME SHALL BE REPORTED TO L&T DESIGN TEAM.
7. SOIL CHARACTERISTICS : SAND WITH SILT AND GRAVEL / GRAVEL WITH SILT AND SAND IS ASSUMED

INVERTER COORDINATES		
POINTS	EASTING	NORTHING
INV-01	263884.549	2506221.334
INV-02	263897.351	2506229.334
INV-03	263909.128	2506237.334
INV-04	263907.722	2506245.334
INV-05	263913.582	2506253.334
INV-06	263915.566	2506253.334
INV-07	263912.591	2506261.334
INV-08	263915.565	2506269.334
INV-09	263913.521	2506269.334
INV-10	263913.241	2506277.332
INV-11	263915.265	2506277.332
INV-12	263915.696	2506285.334
INV-13	263913.672	2506285.334
INV-14	263912.624	2506293.334
INV-15	263915.294	2506301.334
INV-16	263913.170	2506301.334
INV-17	263913.190	2506309.334
INV-18	263913.190	2506317.334
INV-19	263913.190	2506325.334
INV-20	263915.294	2506325.334
INV-21	263913.210	2506333.334
INV-22	263913.310	2506341.334
INV-23	263913.310	2506349.334
INV-24	263915.314	2506349.334
INV-25	263913.190	2506357.334
INV-26	263913.290	2506365.334
INV-27	263913.290	2506373.334
INV-28	263913.190	2506381.334
INV-29	263913.290	2506397.334
INV-30	263913.266	2506405.334
INV-31	263915.294	2506405.334
INV-32	263913.190	2506421.334
INV-33	263994.570	2506469.321
INV-34	263994.470	2506461.334
INV-35	263991.454	2506453.334
INV-36	263991.454	2506445.334
INV-37	263991.434	2506437.334
INV-38	263991.454	2506429.334
INV-39	263991.454	2506421.334
INV-40	263991.454	2506405.334
INV-41	263991.454	2506397.334
INV-42	263992.466	2506397.334
INV-43	263991.454	2506389.334
INV-44	263992.947	2506381.334
INV-45	263993.974	2506373.334
INV-46	263994.470	2506365.334
INV-47	263994.470	2506357.334
INV-48	263994.518	2506349.334
INV-49	263994.470	2506341.334
INV-50	263994.470	2506333.334
INV-51	263994.570	2506325.334
INV-52	263994.507	2506317.334
INV-53	263994.470	2506309.334
INV-54	263994.470	2506301.334
INV-55	263988.312	2506285.341
INV-56	264023.290	2506289.397
INV-57	264023.289	2506297.397
INV-58	264023.290	2506305.404
INV-59	264023.290	2506313.397
INV-60	264024.504	2506321.397

INVERTER COORDINATES		
POINTS	EASTING	NORTHING
INV-61	264023.290	2506329.397
INV-62	264023.290	2506337.397
INV-63	264024.249	2506345.397
INV-64	264024.485	2506353.397
INV-65	264024.298	2506361.397
INV-66	264024.484	2506369.397
INV-67	264024.298	2506377.397
INV-68	264024.484	2506385.397
INV-69	264024.484	2506393.397
INV-70	264024.470	2506401.397
INV-71	264024.484	2506409.397
INV-72	264024.484	2506417.397
INV-73	264024.484	2506425.397
INV-74	264023.492	2506433.397
INV-75	264023.472	2506441.397
INV-76	264023.492	2506449.397
INV-77	264024.390	2506457.397
INV-78	264023.472	2506465.397
INV-79	264023.472	2506473.397
INV-80	264023.472	2506481.397
INV-81	264024.484	2506489.397
INV-82	264023.472	2506497.397
INV-83	264024.113	2506505.397
INV-84	264026.322	2506505.397
INV-85	264028.050	2506513.397
INV-86	264039.676	2506521.397
INV-87	264052.499	2506529.397
INV-88	264065.322	2506537.397
INV-89	264144.114	2506489.444
INV-90	264144.710	2506481.397
INV-91	264144.657	2506465.397
INV-92	264144.610	2506457.397
INV-93	264144.610	2506449.397
INV-94	264144.610	2506441.397
INV-95	264144.710	2506433.397
INV-96	264144.710	2506425.397
INV-97	264144.710	2506417.397
INV-98	264144.710	2506409.404
INV-99	264145.226	2506401.397
INV-100	264143.102	2506401.397
INV-101	264144.610	2506393.397
INV-102	264146.218	2506385.397
INV-103	264144.610	2506385.397
INV-104	264144.622	2506377.397
INV-105	264144.710	2506369.397
INV-106	264146.218	2506361.397
INV-107	264144.114	2506361.397
INV-108	264146.237	2506353.397
INV-109	264144.114	2506353.397
INV-110	264144.114	2506345.397
INV-111	264146.218	2506345.397
INV-112	264144.610	2506337.397
INV-113	264146.218	2506329.397
INV-114	264144.715	2506329.397
INV-115	264146.218	2506321.397
INV-116	264144.114	2506321.397
INV-117	264144.610	2506313.397
INV-118	264144.114	2506305.397
INV-119	264145.206	2506305.397
INV-120	264144.710	2506297.404

ACB COORDINATES		
POINTS	EASTING	NORTHING
ACB-01	263917.148	2506253.168
ACB-02	263916.590	2506261.168
ACB-03	263916.828	2506277.167
ACB-04	263917.649	2506301.168
ACB-05	263917.816	2506325.168
ACB-06	263916.945	2506349.168
ACB-07	263916.753	2506381.168
ACB-08	263916.836	2506397.168
ACB-09	263916.773	2506405.168
ACB-10	263997.065	2506461.168
ACB-11	263992.913	2506421.168
ACB-12	263992.913	2506389.168
ACB-13	263995.891	2506357.168
ACB-14	263996.131	2506325.168
ACB-15	263996.033	2506309.168
ACB-16	264023.359	2506321.231
ACB-17	264023.284	2506353.231
ACB-18	264023.276	2506369.231
ACB-19	264023.281	2506393.231
ACB-20	264023.281	2506425.231
ACB-21	264023.281	2506457.231
ACB-22	264023.281	2506489.231
ACB-23	264026.897	2506513.231
ACB-24	264142.898	2506481.231
ACB-25	264143.306	2506465.231
ACB-26	264143.142	2506433.231
ACB-27	264143.260	2506409.231
ACB-28	264143.122	2506385.231
ACB-29	264143.097	2506369.231
ACB-30	264143.097	2506345.231
ACB-31	264143.101	2506321.231
ACB-32	264143.097	2506305.231

ETHERNET BOX- COORDINATES		
POINTS	EASTING	NORTHING
ETH-01	263993.032	2506357.199
ETH-02	263993.032	2506309.199
ETH-03	263911.766	2506325.199
ETH-04	263911.732	2506301.199
ETH-05	263912.590	2506269.199
ETH-06	263907.689	2506237.199
ETH-07	263911.752	2506357.199
ETH-08	263911.827	2506405.199
ETH-09	263990.016	2506397.199
ETH-10	263990.016	2506445.199
ETH-11	264023.122	2506361.263
ETH-12	264024.229	2506313.263
ETH-13	264145.053	2506361.263
ETH-14	264146.049	2506337.263
ETH-15	264146.049	2506313.263
ETH-16	264146.049	2506393.263
ETH-17	264146.149	2506425.263
ETH-18	264146.096	2506465.263
ETH-19	264023.204	2506401.263
ETH-20	264024.381	2506449.263
ETH-21	264024.361	2506497.263
ETH-22	264040.564	2506521.262

DBB-COORDINATES		
POINTS	EASTING	NORTHING
DBB-01	263911.639	2506341.199
DBB-02	263991.352	2506373.199
DBB-03	264026.100	2506377.263
DBB-04	264146.386	2506377.263

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		D01	09/03/2023	AS-BUILT	RVJ	AVG	GSK	SJM	
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DRAWN	RVJ	23/08/2020
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SCALE	NTS	A1

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